Positive Reference-Modular Localization and Bounds for the Wilson Majorant Jacobi Moment

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Abstract

For the half-filled finite sine-kernel reference and a Wilson positive majorant, we prove a parameterized spectral-wing bound for the positive reference-modular trace and derive complementary upper and lower logarithmic bounds for the positive majorant-weighted Jacobi moment. The arguments use finite-dimensional order inequalities, the established Wilson majorant tail, the exact signed Jacobi first moment, and a published discrete-prolate eigenvalue bound. No uniform finite-size approximation of the reference modular generator by the Jacobi generator is assumed. The full fixed-positive-scaled-mass entropy expansion is proved separately in Theorem 1 of *Finite Part of Wilson–Dirac Relative Entropy*, with remainder $O_\mu(1/\log L)$. Theorem 3 of *Uniform Modular Control and Nonlinear Convergence for Wilson–Dirac Intervals* proves a finite-mode bound $\|E_N\| \le 7$; it does not assert a uniform moving-edge expansion.

1. Operators and mathematical inputs

Let $N = 2L$ be even and $\mu > 0$ fixed. The reference covariance and commuting Jacobi matrix are

$$B_N(i,j) = \begin{cases} \frac{1}{2} & i = j, \\ \frac{\sin[\pi(i-j)/2]}{\pi(i-j)} & i \neq j, \end{cases} \qquad (J_N)_{j-1,j} = \frac{j(N-j)}{N},$$

with all other Jacobi entries zero. The reference modular operator and the positive Wilson majorant are

$$K_N = \log(I - B_N) - \log B_N, \qquad G_N \succeq 0, \qquad M_N = \operatorname{Tr} G_N \le \frac{\mu}{\pi}.$$

Write $S_N = S_N^*$ for the longitudinal Wilson covariance perturbation and assume the exact operator inequalities $-G_N \preceq S_N \preceq G_N$. The following established majorant tail is used as a mathematical input, for every $t > 0$:

$$\nu_N((t,\infty)) := \operatorname{Tr}\big(G_N \mathbf{1}_{\{|J_N| > t\}}\big) \le \frac{a_\mu}{t} + \frac{b_\mu}{t^2}, \quad a_\mu = \sqrt{\frac{2}{15}}\frac{\mu^2}{\pi}, \quad b_\mu = \frac{\mu}{3\pi}. \tag{1}$$

The exact longitudinal first moment has the asymptotic

$$\operatorname{Tr}(S_N J_N) = -\frac{\mu^2}{6\pi}\log N + \frac{\mu^2}{4\pi} - \frac{\mu^2}{6\pi}\log\frac{4}{\mu} + O_\mu\Big(\frac{\log N}{N}\Big). \tag{2}$$

The published discrete-prolate lower eigenvalue estimate, combined with half-filled particle–hole symmetry, gives

$$\|K_N\| \le 2N \operatorname{arsinh}(1) + 3\log N + C \tag{3}$$

for an absolute C and even N. The full Wilson entropy is already reduced, under its separately established hypotheses, to

$$\mathcal{D}_L(\mu) = P_N + C_\mu + O_\mu(\sqrt{\log N}), \quad P_N = \mathrm{Tr}(G_N|K_N|), \quad C_\mu = \frac{\mu^2}{4} - \frac{\mu^2}{6}\log\frac{8}{\mu}. \tag{4}$$

The explicit C_μ in (4) is **not** a certified entropy finite part: the displayed remainder has not been shown to converge. Equations (1)–(4) are inputs from the cited studies, not independently re-proved in this manuscript.

2. A spectral-wing trace inequality

Lemma 1 (order domination). Let $G \succeq 0$, $-G \preceq S = S^* \preceq G$, and $F = F^*$ be finite matrices. Then

$$\boxed{\left|\mathrm{Tr}(SF)\right| \le \mathrm{Tr}(G|F|).} \tag{5}$$

Proof. Decompose $F = F_+ - F_-$ spectrally, with $F_\pm \succeq 0$. Trace monotonicity against each positive part gives $\left|\mathrm{Tr}(SF_\pm)\right| \le \mathrm{Tr}(GF_\pm)$. The triangle inequality yields (5). No commutativity between G and F is required. □

Let $\Pi_{>R} = \mathbf{1}_{\{|J_N|>R\}}$, $\Pi_{\le R} = I - \Pi_{>R}$, and $P_N^{\le R} = \mathrm{Tr}(G_N|K_N|\Pi_{\le R})$. Because $[B_N, J_N] = 0$, the operators $|K_N|$ and $\Pi_{>R}$ commute and their product is positive.

Theorem 2 (quantitative positive-trace localization). Under (1) and (3), for all $R > 0$,

$$\boxed{0 \le P_N - P_N^{\le R} \le \left(2N\,\mathrm{arsinh}(1) + 3\log N + C\right)\left(\frac{a_\mu}{R} + \frac{b_\mu}{R^2}\right).} \tag{6}$$

Proof. Positivity follows from $\mathrm{Tr}(G_N X) \ge 0$ whenever $G_N, X \succeq 0$, taking $X = |K_N|\Pi_{>R}$. Since $0 \preceq X \preceq \|K_N\|\Pi_{>R}$, trace monotonicity, (1), and (3) imply (6). □

Corollary 3 (adjustable sublogarithmic window). Let $h_N \to \infty$ with $h_N = o(\log N)$, and set $R_N = N/h_N$. Then

$$\boxed{P_N - P_N^{\le R_N} = O_\mu(h_N) = o_\mu(\log N),} \tag{7}$$

and (4) becomes

$$\boxed{\mathcal{D}_L(\mu) = P_N^{\le N/h_N} + C_\mu + O_\mu(\sqrt{\log N} + h_N).} \tag{8}$$

Proof. Substitute $R_N = N/h_N$ in (6). The leading term is $2a_\mu\,\mathrm{arsinh}(1)h_N$, with lower-order contributions $O_\mu(h_N \log N/N + h_N^2/N)$, hence (7). Equation (8) follows from (4). □

For example, $h_N = \log N/\log\log N$ produces the narrower region $|J_N| \le N\log\log N/\log N$ while the discarded wing remains $o(\log N)$. This choice **worsens the stated remainder** relative to $h_N = \sqrt{\log N}$; it only tightens the window relevant for a leading-coefficient proof. In general the coarse bound requires $R_N \gg N/\log N$. This is a sufficient scale for this method, not a necessary physical threshold.

The desired complete leading logarithmic coefficient is equivalent, under (4), to

$$\boxed{P_N^{\le N/h_N} = \frac{\mu^2}{6}\log N + o_\mu(\log N).} \tag{9}$$

Equation (9) follows by combining the separate leading entropy law and positive-trace reduction with the discarded-wing estimate above. The finite-dimensional wing bound alone does not determine its coefficient.

3. Positive majorant-weighted Jacobi moment

Define $Q_N(\mu) = \operatorname{Tr}(G_N|J_N|)$. Since J_N is a self-adjoint Jacobi matrix and $\|J_N\| \le N/2$, the layer-cake representation gives

$$Q_N(\mu) = \int_0^{N/2} \nu_N((t,\infty))\, dt. \tag{10}$$

Theorem 4 (logarithmic coefficient bracket). For every fixed $\mu > 0$,

$$\boxed{\frac{\mu^2}{6\pi} \le \liminf_{N\to\infty,\ N\,\text{even}} \frac{Q_N(\mu)}{\log N} \le \limsup_{N\to\infty,\ N\,\text{even}} \frac{Q_N(\mu)}{\log N} \le \sqrt{\frac{2}{15}}\frac{\mu^2}{\pi}.} \tag{11}$$

Proof. Lemma 1 with $F = J_N$ gives $Q_N \ge |\operatorname{Tr}(S_N J_N)|$. The exact asymptotic (2) then proves the lower limit. For the upper limit, split (10) at $t = 1$. For $N \ge 2$, the interval $(0,1)$ contributes at most M_N, while (1) gives

$$Q_N \le M_N + a_\mu \log(N/2) + b_\mu(1 - 2/N).$$

Divide by $\log N$ and let $N \to \infty$, using $M_N \le \mu/\pi$. □

The lower bound already has the coefficient $\mu^2/(6\pi)$ anticipated for Q_N, but the majorant-tail estimate alone does not establish a matching upper bound. A numerical observation that $Q_N + \operatorname{Tr}(S_N J_N)$ remains small over tested finite sizes is not a proof of boundedness or decay.

4. A sufficient two-part proof obligation

The positive-trace problem can be decomposed *algebraically* as

$$P_N = \pi Q_N + \underbrace{\operatorname{Tr}\big[G_N(|K_N| - \pi|J_N|)\big]}_{\Delta_N^{\mathrm{abs}}}. \tag{12}$$

Therefore the pair of estimates

$$Q_N = \frac{\mu^2}{6\pi}\log N + o_\mu(\log N), \qquad \Delta_N^{\mathrm{abs}} = o_\mu(\log N) \tag{13}$$

would imply (9), and hence the full leading relative-entropy coefficient. These conditions are sufficient, **not individually necessary**, because cancellations between two terms cannot be ruled out. Neither estimate in (13) is proved here.

The positive operator structure and signed exact Jacobi moment thus give a precise framework for separating majorant spectral-weight growth from the modular comparison. The separate weighted theorem in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law* gives $P_N = \pi Q_N + O_\mu(1)$; combined with the proved entropy law and the positive-trace reduction, this also yields $Q_N = (\mu^2/(6\pi))\log N + O_\mu(\sqrt{\log N})$.

5. Scientific limitations and references

All operator statements concern the infinite-chain occupied-band projectors compressed to the specified finite interval. The positive-majorant tail, entropy reduction, and elliptic first moment are mathematical inputs from the named companion papers. Numerical experiments on finite matrices are corroboration only. Fixed-index discrete-prolate asymptotics must not be substituted for estimates uniform over moving spectral indices. The finite-part convergence and any uniform moving-edge expansion are separate from the established leading entropy coefficient.

References

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